> *This hands-on exercise demonstrates the heart of Laravel's MVC architecture. You're now ready to build more complex pages with routing, logic, and templating.*

3.8 Exercise 3: Creating a Laravel 12 Web Calculator

3.8.1 Description

In this lab, we will create a simple calculator web app using Laravel 12. The calculator will take two numbers and an operator (add, subtract, multiply, divide), then display the result. This project demonstrates Laravel's MVC architecture with form input, routing, controllers, and Blade views.

3.8.2 Objectives

By the end of this lab, you will be able to:

- Handle form submissions in Laravel
- Use routes and controllers to process user input
- Pass and display calculated results in Blade views
- Understand MVC interaction through a practical scenario

3.8.3 Prerequisites

Ensure the following are installed: - PHP ≥ 8.2 - Composer - Laravel 12 - Visual Studio Code - Basic terminal usage

> *This lab uses **PHP 8.4**, **Laravel 12** and **Visual Studio Code** as the primary editor.*

3.8.4 Steps

Here's a step-by-step guide to creating a simple calculator web app using Laravel 12:

3.8.4.1 Step 1: Create and Open Laravel Project